Quasi-Brody Hyperbolicity for Non-normal Crossing Divisors

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Lang-Vojta Conjecture

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Lang-Vojta (Bombieri–Lang) Conjecture

Let k be a number field and let S be a finite set of places of k. Let (X, D) be a pair defined over k. If (X, D) is of log general type, the any set of (D, S)-integral points in $X(k)$ is contained in a proper Zariski closed subset of X.

Brody Hyperbolicity

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- For the purpose of this talk, we define the quasi-Brody hyperbolicity.

Quasi-Brody Hyperbolicty

Let X be a complex analytic space. We say X is quasi-Brody hyperbolic if the image of every nonconstant holomorphic maps $f : \mathbb{C} \to X$ is contained in a proper algebraic subset $Z \subsetneq X$.

Green-Griffiths-Lang Conjecture

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Green-Griffiths-Lang Conjecture

Let X be a smooth complex projective variety, and let D be a normal crossings divisor on X. Suppose that $X \setminus D$ is a variety of log-general type. Then:

1. Every entire curve $f : \mathbb{C} \to X \setminus D$ is algebraically degenerate; that is, the image of f is contained in a proper algebraic subvariety of X.
2. Furthermore, there exists a proper algebraic subvariety $Z \subset X$ such that the image of every entire curve $f : \mathbb{C} \to X$ is contained in Z.

Quasi-Brody Hyperbolicity

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Theorem(Wang-X.)

Let $\dim X = n$, and $D_1, \ldots, D_{n+1}$ effective Cartier divisors such that they are all numerically equivalent to an ample divisor A. Assume that any n-tuple of divisors among $D_1, \ldots, D_{n+1}$ intersect properly, and that $\bigcap_{i=1}^{n+1} D_i \neq \emptyset$. Suppose further that for every point $Q \in \bigcap_{i=1}^{n+1} D_i$ and every index set I of n elements, the following inequality holds:

$$(D_i)_Q^I < \left(\frac{n}{n+1}\right)^n D_i^I$$

Then $X \setminus \bigcup_{i=1}^{n+1} D_i$ is quasi-Brody hyperbolic.

Quasi-Brody Hyperbolicity

- This result complements that of Noguchi, Winkelmann, and Yamanoi concerning the Green-Griffiths-Lang conjecture for $X = \mathbb{P}^n$ with D consisting of $n+1$ normal crossings hypersurfaces.

Quasi-Brody Hyperbolicity

- This result complements that of Noguchi, Winkelmann, and Yamanoi concerning the Green-Griffiths-Lang conjecture for $X = \mathbb{P}^n$ with D consisting of $n+1$ normal crossings hypersurfaces.
- Different from classical results in several complex variables, we do not assume log-general type a priori. And the proving techniques are from Diophantine approximation.

An Example

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Example

Let $X = \mathbb{P}^2$, let

$$D_1 : x = 0, D_2 : y = 0, D_3 : x^2 + y^2 + xz = 0.$$

Let $Q = [0 : 0 : 1]$. In this case, we have $2D_1 \equiv 2D_2 \equiv D_3$ and

$$(D_1.D_3)_Q = (D_1.D_3) = 2.$$

This example does not satisfy the condition on intersection numbers. Now consider the entire curve:

$$f = (1, e^z, e^{z^2} - e^{2z} - 1) : \mathbb{C} \to \mathbb{P}^2 \setminus D_1 \cup D_2 \cup D_3.$$

Clearly, f is algebraically non-degenerate.

The End

Thank you!